

Name: _____

Student ID: _____

ECON 0150 | MiniExam 5 | Spring 2026

This MiniExam will take 20 minutes with a quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material. Answer clearly, completely, and concisely.

Academic Conduct Code

The following academic conduct code is designed to protect the integrity of your work. Print your name/initials beside the three academic honesty agreements. I pledge to my fellow students, the university, and the instructor, that:

____ I will complete this MiniExam solely using my own work.

____ I will not use any digital resources unless explicitly allowed by the instructor.

____ I will not communicate directly or indirectly with others during the MiniExam.

Q1. An economist studies how minimum wage affects state employment, holding cost of living constant.

a) Which regression model answers this question? (circle one)

A) $\text{employment} = \beta_0 + \beta_1 \cdot \text{min_wage} + \varepsilon$

B) $\text{employment} = \beta_0 + \beta_1 \cdot \text{cost_of_living} + \varepsilon$

C) $\text{employment} = \beta_0 + \beta_1 \cdot \text{min_wage} + \beta_2 \cdot \text{cost_of_living} + \varepsilon$

D) $\text{employment} = \beta_0 + \beta_1 \cdot \text{min_wage} + \beta_2 \cdot (\text{min_wage} \times \text{cost_of_living}) + \varepsilon$

b) Interpret the coefficient on `min_wage` in one sentence:

Q2. A health economist wants to test whether counties that adopted Medicaid expansion have different uninsured rates than counties that did not, while controlling for the county poverty rate. The variable `expansion` is coded as 1 for counties that adopted expansion and 0 for those that did not.

a) Write the regression model for this question:

_____ = $\beta_0 + \beta_1 \cdot$ _____ + $\beta_2 \cdot$ _____ + ε

b) Interpret the coefficient on `expansion` in one sentence:

c) Why might we want to control for poverty rate when studying the effect of Medicaid expansion on the uninsured rate?

Q3. A researcher collected data on 150 used cars and estimated the following model:

$$\text{price} = \beta_0 + \beta_1 \cdot \text{mileage} + \beta_2 \cdot \text{accident} + \varepsilon$$

Where price is in dollars, mileage is in thousands of miles, and accident = 1 for cars with an accident history and accident = 0 for those without. The regression output shows:

	coef	std err	t	P> t
Intercept	22000.000	850.000	25.882	0.000
mileage	-120.000	14.000	-8.571	0.000
accident	-3200.000	560.000	-5.714	0.000

a) What is the predicted price for a car with 60,000 miles and no accident history?

Answer: \$_____

b) What is the predicted price for a car with 60,000 miles and an accident history?

Answer: \$_____

c) Interpret the coefficient on accident (-3200) in one sentence:

Q4. A school estimates how study hours affect exam scores, allowing the effect to differ based on whether the student received tutoring (1 for tutored; 0 for not tutored).

$$\text{score} = \beta_0 + \beta_1 \cdot \text{hours} + \beta_2 \cdot \text{tutor} + \beta_3 \cdot (\text{hours} \times \text{tutor}) + \varepsilon$$

The estimated coefficients are: $\beta_0 = 65$, $\beta_1 = 2.5$, $\beta_2 = 4$, $\beta_3 = 1.5$.

a) What is the effect of one additional hour of studying on exam scores for students who were not tutored?

Answer: _____

b) What is the effect of one additional hour of studying on exam scores for students who were tutored?

Answer: _____

c) The interaction coefficient ($\beta_3 = 1.5$) tells us: (circle one)

- A) Tutored students have 1.5 higher exam scores on average
- B) An additional hour of studying increases exam scores by 1.5 more for tutored students than non-tutored students
- C) Studying is 1.5 times as effective for tutored students
- D) An additional hour of studying increases exam scores by 1.5 points for tutored students